UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
Department of Physics and Astronomy

Standing Waves

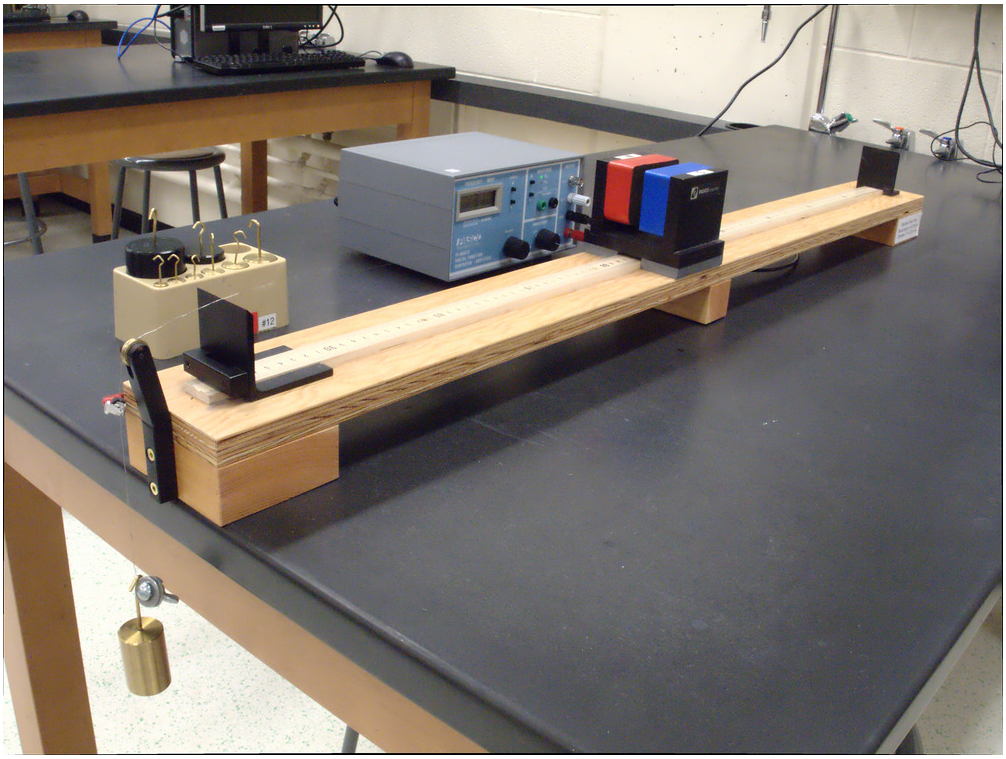


Figure 1.1. Standing waves experimental setup.

Equipment: set of connecting leads, Pasco digital function amplifier, set of hanging masses, standing waves wire apparatus

Goals of the Experiment: To test the prediction, within experimental uncertainty, that the difference between any two successive frequencies of standing waves on a string under tension should be the same and that the propagation speed should be proportional to the square root of the tension force.

Theory

Background

A string under tension and fixed at both ends can sustain transverse **standing waves** of the form

$$y(x,t) = A \sin \omega t \sin kx \tag{1.1}$$

where $x = 0$ at one end, and $x = L$ at the other end, ω is the angular frequncy and k is the wavenumber. In the lectures, you will have seen that the possible wavelengths are

$$\lambda_n = \frac{2L}{n} \tag{1.2}$$

for $n = 1, 2, 3, \ldots$. In this way, $y(0,t) = y(L,t) = 0$ always, which is consistent with the two ends being fixed. Furthermore, since the propagation speed $v = f\lambda$, the frequencies are also limited to the set

$$f_n = n\frac{v}{2L} \tag{1.3}$$

In the case of a string under tension, if the tension force is T and the mass per unit length of the string is μ, the propagation speed is

$$v = \sqrt{\frac{T}{\mu}} \tag{1.4}$$

This is the basis for all stringed instruments. A violinist, for example, excites a standing wave in a string by running the bow over it, which actually grips the string alternately in what is called a **stick-slip cycle**. In this lab, we will produce standing waves in a wire not with a bow, but by an method, involving an alternating current and a magnetic field.

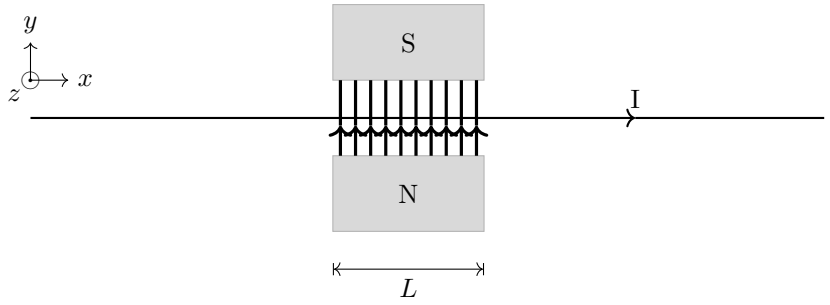


Figure 1.2. Exciting a string by means of an alternating current and a magnetic field. The string passes between the poles of a magnet that produces a uniform magnetic field in the $+y$ direction. A current flows in the $+x$ direction. As a result, a force $\vec{F}$ is exerted on the wire, in the $+z$ direction (not shown).

Figure 1.2 shows the mechanism schematically. A current I flows along the wire in the $+x$ direction. A segment of length L is between the poles of a permanent magnet, with magnetic field $B\hat{\jmath}$. The force on the wire is

$$\vec{F} = I(L\hat{\imath}) \times (B\hat{\jmath}) = ILB\hat{k} \tag{1.5}$$

If we send an alternating current $I = I_0 \cos \omega t$ through the wire, there will be a transverse oscillating force with the same frequency on the wire, forcing it to oscillate at that frequency. If that happens to be one of the normal mode frequencies according to 1.3, then a large amplitude standing wave is generated. Otherwise the induced wave dies down very quickly, and the continued forcing only produces a vibration of negligible amplitude. By sweeping through a range of frequencies, we should then be able to observe the spectrum of normal modes within that range.

In this lab, we are going to test two predictions, to within the experimental uncertainty:

- the difference between any two successive frequencies should be the same (Equation 1.3), and
- the propagation speed should be proportional to the square root of the tension force (Equation 1.4).

The Experiment

Figure 1.3 shows the experimental setup. The string lies across two black metal plates with knife edges, with one end fixed to the platform and the other wrapped over a pulley and a weight attached to it. A ruler running down the centre of the platform allows length measurements. The structure in the middle is a magnet with the north and south poles painted red and blue. The wire passes between the poles of the magnet, which can be displaced freely along the ruler.

A Pasco signal generator creates a sinusoidal voltage that is applied to the ends of the wire. The frequency of the alternating voltage can be varied by simply turning a knob on the signal generator.

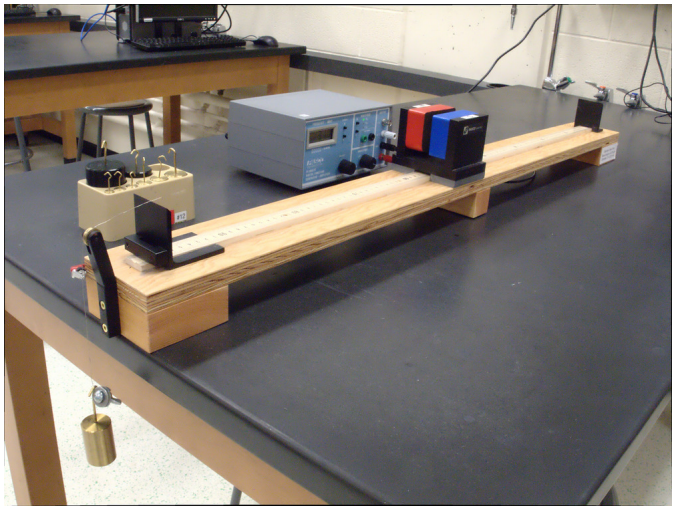


Figure 1.3. The experimental setup.

Experiment 1: Generating harmonics for a given value of the tension in the wire

Hang a 500 g mass from the free end of the wire. You will use the signal generator to excite successive harmonics in the wire, starting with the fundamental mode, for which the entire length of the wire is half a wavelength. Begin by calculating the frequency of the fundamental mode using Equation 1.3 with $n = 1$, and with the speed given by 1.4. Notice that the mass per unit length is provided in the Introductory Presentation.

- Record the length of the wire between the two knife edges (use the ruler). Estimate the uncertainty in the length of the wire.
- Record the value of the mass. Since the uncertainty in the mass is much less than the other uncertainties in this experiment, it may be considered exact.
- Record the value of the mass per unit length, μ, with its uncertainty.

Place the magnet at the middle of the wire (later you may move it away in order to see the nodes better and to prevent the wire from touching the magnet). Turn down the amplitude knob on the signal generator to zero, and turn on the power. Then tune the frequency to somewhat below your estimated frequency, turn up the amplitude a bit, and slowly sweep upwards in frequency. You should see the wire vibrate clearly in the fundamental mode when you hit the right frequency.

- Record the frequency of the fundamental mode, f_1. Also, experiment a little with increasing and decreasing the frequency to estimate the uncertainty, and record this estimate as well.

In order to estimate the uncertainty in f_n , find the difference between the frequency when the amplidude of the stanading wave is at its maximum and when the amplitude is reduced to half of its maxiumum value.

Next, you will excite the second harmonic. Please note that you will need to find an optimal position for the magnet.

Now sweep up in frequency again, until you excite the second harmonic.

- Record f_2 with its uncertainty.

Continue in this way measuring the frequencies of the first twelve harmonics.

- For each n from 1 to 12, record f_n, with its uncertainty.

Hint: for higher harmonics the amplitude is lower and you might not be able to see them clearly, but you could listen for the harmonics. Estmate the uncertainty only for the harmonics you will be able to see.

Experiment 2: Measuring the dependence of v on tension, T

This time, instead of keeping the tension constant and varying the frequency, you will vary the tension by hanging different weights from the string, and measure the frequency of the fundamental mode each time. The idea is to calculate v from f_1 using Equation 1.3, and see how it varies with T. Repeat for at least 6 masses, increasing the mass by 100 g each time.

- For at least 6 masses, record the value of the mass, and the frequency of the fundamental mode with its uncertainty.

Uncertainty Estimation and Propagation

There are many uncertainty calculations in the data analysis of this lab. In your report, you should show how the uncertainty propagation formula was applied. For example, if $v = \sqrt{T/\mu}$, then the uncertainty in v is

$$\begin{aligned} u(v) &= \sqrt{\left(\frac{\partial v}{\partial T}\right)^2 u^2(T) + \left(\frac{\partial v}{\partial \mu}\right)^2 u^2(\mu)} \\ &= \sqrt{\frac{1}{4T\mu}u^2(T) + \frac{T}{4\mu^3}u^2(\mu)} \end{aligned} \tag{1.6}$$

Since $u(T)$ is negligible, this reduces to

$$u(v) = \frac{1}{2}\sqrt{\frac{T}{\mu^3}}u(\mu) \tag{1.7}$$

Pre-Lab

Your Pre-lab exercises are available on the D2L Lab site (Assessment – Quizzes) and should be completed there.

Data analysis

Experiment 1

Use the Excel Line fitting template to fit a straight line to your measured values of f_n versus n (use the option allowing you to take the uncertainties of individual measurements into account). Record the value of the slope with its uncertainty. Separately, calculate the speed as $v = \sqrt{T/\mu}$, and use the measured value of L to calculate the expected value of the slope. Then compare the experimental value of the slope with the expected value to see if they agree within the experimental uncertainty (use the t-test as decribed in the Measurement Uncertainty document).

Experiment 2

If the relationship between v, T, and μ is $v = (T/\mu)^{1/2}$, then taking the logarithm of both sides gives a linear relationship between $\ln v$ and $\ln T$.

For each mass in Experiment 2, calculate the tension in the string, and the speed from the frequency and the length of the string using Equation 1.3, with its uncertainty[1]. Then calculate $\ln T$, and $\ln v$ with its uncertainty. Use the Excel Line fitting template to fit straight line to these data pairs, and record the value of the slope and their intercept with their uncertainties. Then you can compare these values with the expected values, and see if they agree within the experimental uncertainty.

Lab Report

Your lab report should include the following items:

- A title; your name, and those of your collaborators;
- Objective of the experiment (a summarizing statement about what you set out to observe); Short description of experimental methods;
- Data: recorded values of the length of the wire, and the mass per unit length, with their uncertainties; For experiment 1, the value of the mass. A table with the values of the frequencies recorded, with their uncertainties. For Experiment 2, the recorded values of the masses, and the corresponding fundamental frequencies with their uncertainties.
- Analysis: The graph with the line of best fit to f_n vs. n, and the value of the slope with its uncertainty. The calculated expected value of v with its uncertainty. The expected value of the slope with its uncertainty. A table with the calculated values of $\ln T$ and $\ln v$ with its uncertainty. For Experiment 2, a table with the calculated values of the tension in the wire and the speed and its uncertainty for each mass. Show sample calculations for each table.
- Discussion: The graph with the line of best fit to $\ln v$ vs. $\ln T$, and the values of the slope and intercept with their uncertainties. The calculated expected values of the slope and intercept with their uncertainties.
- Discussion: For Experiment 1 and 2, a comparison of the expected value and the experimental value of the slope, with your conclusion about whether they agree to within the experimental uncertainty; comments on possible sources of errors;
- Your conclusion about whether the two ratios are equal within the experimental uncertainties.

[1]The uncertainty in $T = mg$ is negligible.